

ANOVA Analysis of LLM Model Accuracy

Course Project | STAT 332 - Design of Experiments | Spring 2026

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Data Description

```
data <- read_csv("data.csv")
kable(head(data, 10), caption = "First 10 rows of the dataset")
```

Table 1: First 10 rows of the dataset

batch_size	replicate	model	language	accuracy
1	1	claude	en	1.0
1	2	claude	en	0.0
1	3	claude	en	1.0
1	4	claude	en	1.0
1	5	claude	en	1.0
5	1	claude	en	0.8
5	2	claude	en	1.0
5	3	claude	en	0.8
5	4	claude	en	0.8
5	5	claude	en	1.0

```
# turn the language, model, and batch_size variable into a factor
data <- data %>%
  filter(batch_size != 1) %>% # 0 or 1 score
  mutate(language = as.factor(language),
         model = as.factor(model),
         batch_size = as.factor(batch_size))
```

Table 2: Description of the variables in the dataset

variable	description	type	levels
accuracy	The accuracy of the LLM model on the test set	Response	NA
language	The language used in the prompt	Factor	English, Arabic, Mandarin
model	The LLM model used for prediction	Factor	claude, chatgpt, deepseek, gemini
batch_size	Number of MCQs in the prompt	Factor	1, 5, 10, 20

Exploratory Data Analysis

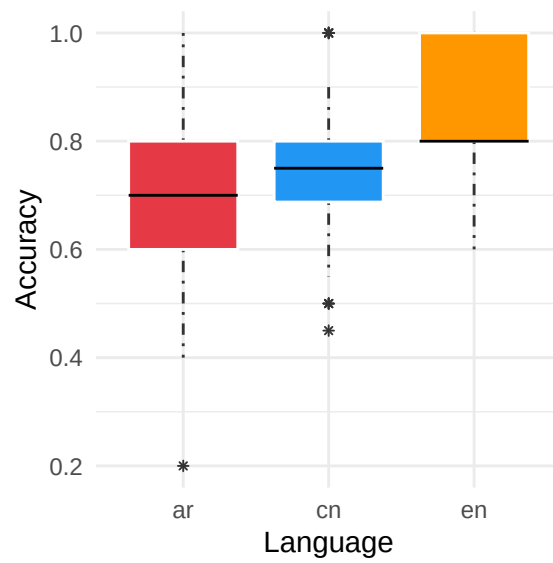


Figure 1: Boxplot of accuracy by language

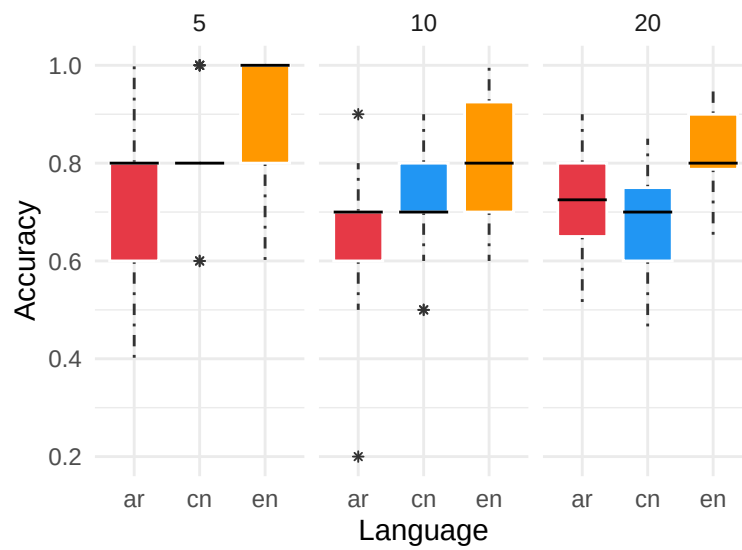


Figure 2: Boxplot of accuracy by language and batch size

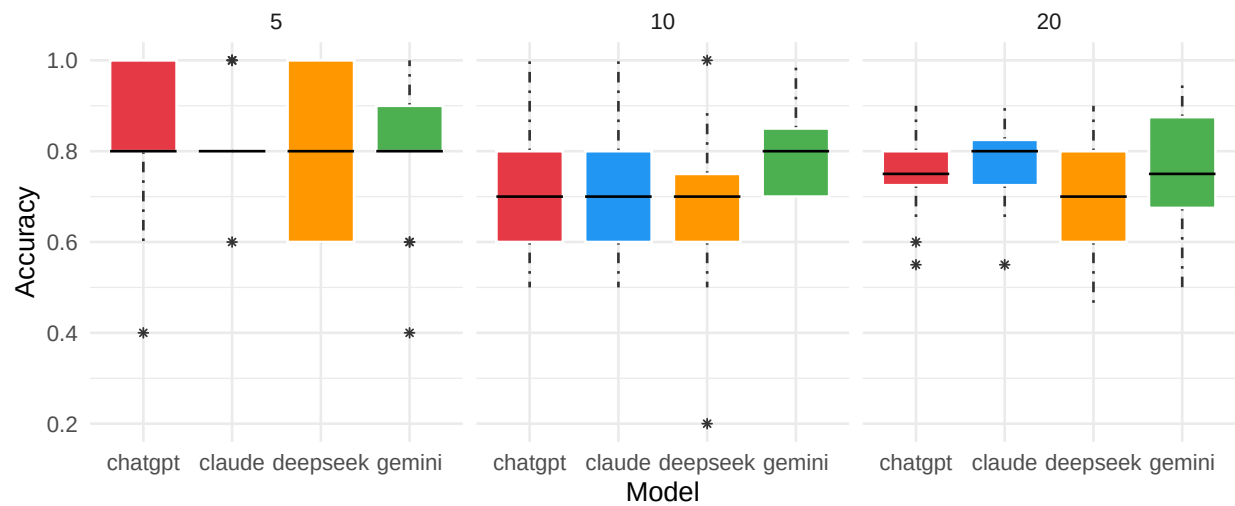


Figure 3: Boxplot of accuracy by model and batch size

Hypothesis Testing

One-way ANOVA

```
# one-way ANOVA for language
fit1 <- aov(accuracy ~ language, data = data)
summary(fit1)
```

```
language      Df Sum Sq Mean Sq F value    Pr(>F)    
Residuals    177  3.0044   0.0170               

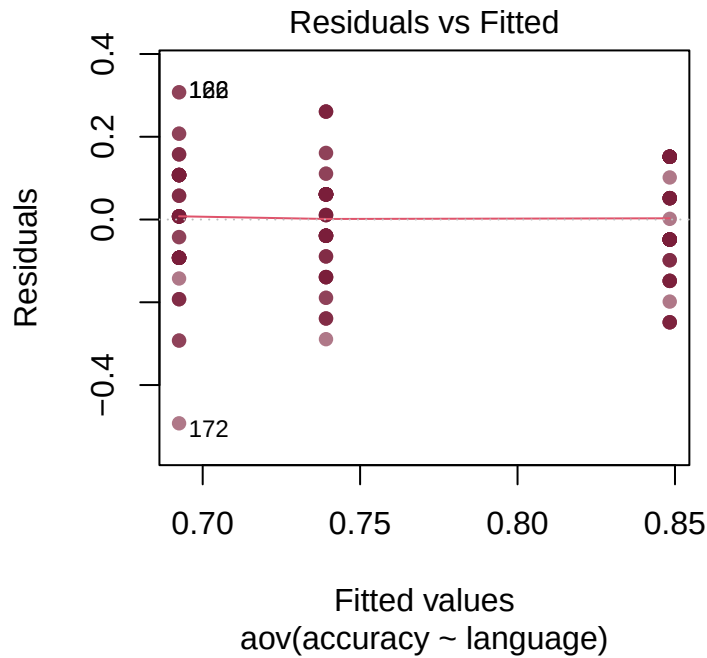
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# language is significant
# check assumptions of ANOVA
# normality
shapiro.test(residuals(fit1))
```

Shapiro-Wilk normality test

```
data: residuals(fit1)
W = 0.97759, p-value = 0.005342
```

```
# not met
# homogeneity of variances
plot(fit1, 1,
     col = adjustcolor("#7A1E3A", 0.6),
     pch = 16) # residuals vs fitted values
```



```
# one-way ANOVA for model
fit2 <- aov(accuracy ~ model, data = data)
summary(fit2)
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
model          3  0.117  0.03915    1.885  0.134
Residuals    176  3.655  0.02076
```

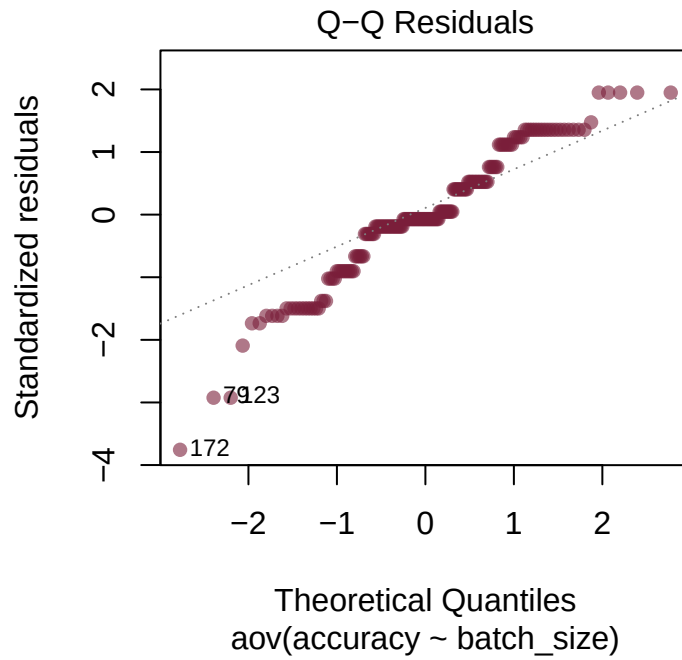
```
# model is not significant
```

```
# one-way ANOVA for batch size
fit3 <- aov(accuracy ~ batch_size, data = data)
summary(fit3)
```

```
              Df Sum Sq Mean Sq F value  Pr(>F)
batch_size     2  0.233  0.11667    5.836 0.00351 **
Residuals    177  3.539  0.01999
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# batch size is significant
```

```
# Normality assumption is not met for batch size
plot(fit3, 2, col = adjustcolor("#7A1E3A", 0.6), pch = 16 ) # normal Q-Q plot
```

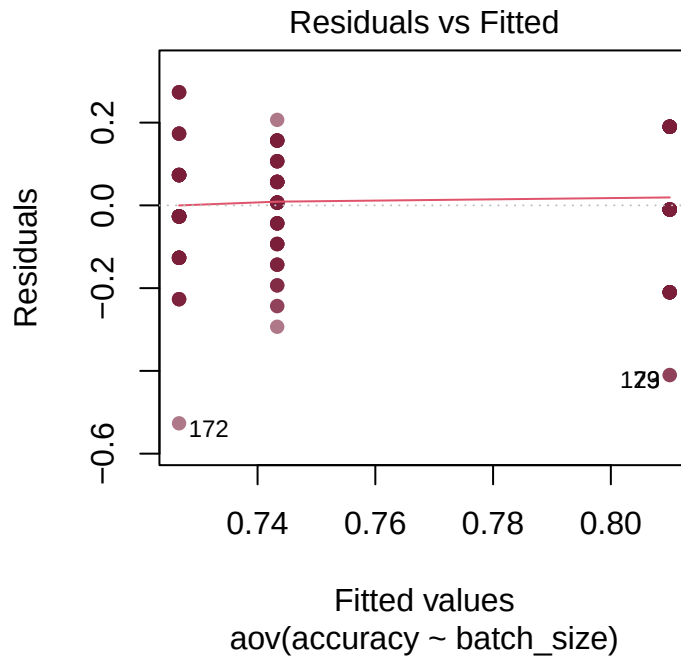


```
shapiro.test(residuals(fit3))
```

Shapiro-Wilk normality test

```
data: residuals(fit3)
W = 0.95717, p-value = 2.73e-05
```

```
# not met
plot(fit3, 1,
     col = adjustcolor("#7A1E3A", 0.6),
     pch = 16) # residuals vs fitted values
```



```
# two-way ANOVA for language and batch size
fit4 <- aov(accuracy ~ language * batch_size, data = data)
summary(fit4)
```

```
              Df Sum Sq Mean Sq F value    Pr(>F)
language         2  0.7676   0.3838   24.782 3.54e-10 ***
batch_size        2  0.2333   0.1167    7.533 0.000732 ***
language:batch_size  4  0.1228   0.0307    1.983 0.099262 .
Residuals       171  2.6482   0.0155
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# language, batch size are significant
# But the interaction is not significant at 0.05 level
```

```
# remove the interaction term and fit the model again
fit5 <- aov(accuracy ~ language + batch_size, data = data)
summary(fit5)
```

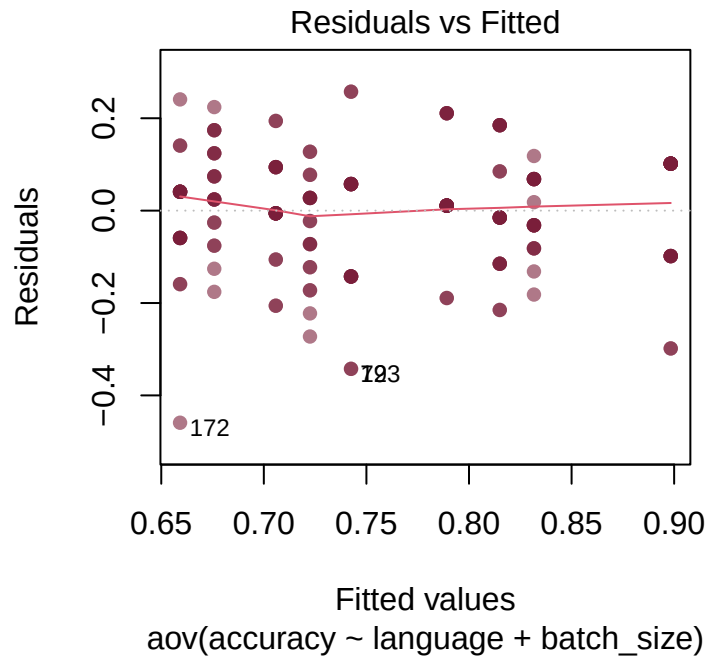
```
              Df Sum Sq Mean Sq F value    Pr(>F)
language         2  0.7676   0.3838   24.237 5.11e-10 ***
batch_size        2  0.2333   0.1167    7.368 0.000847 ***
Residuals       175  2.7711   0.0158
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# check assumptions of ANOVA
shapiro.test(residuals(fit5))
```

Shapiro-Wilk normality test

```
data: residuals(fit5)
W = 0.97761, p-value = 0.00538
```

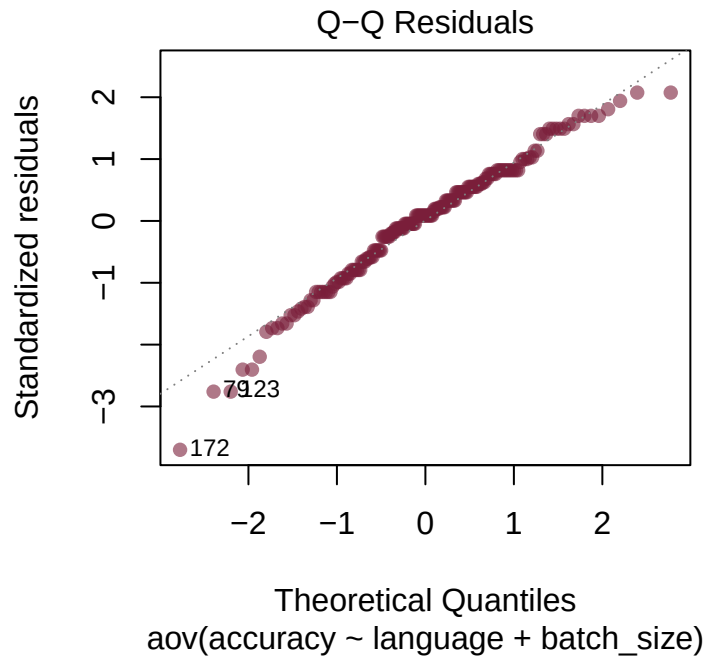
```
# not met, p-value is less than 0.05
plot(fit5, 1,
     col = adjustcolor("#7A1E3A", 0.6),
     pch = 16) # residuals vs fitted values
```



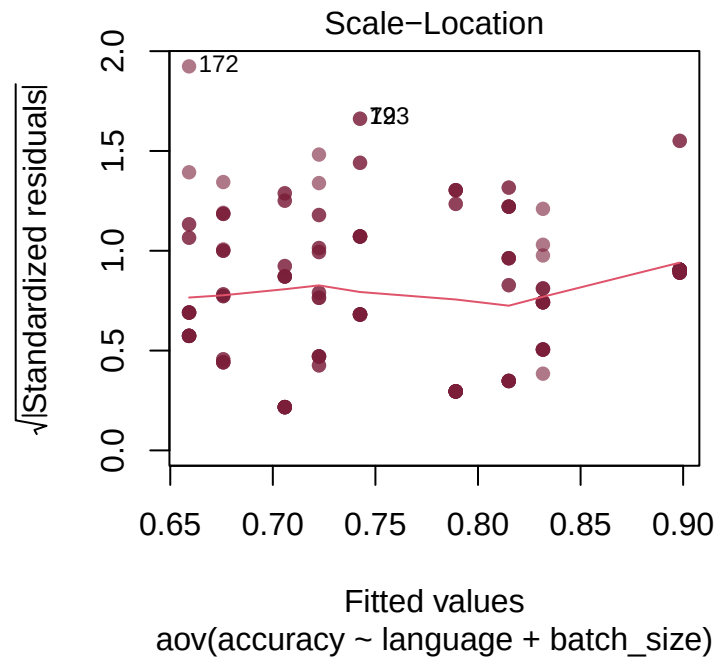
```
# p-value is more than 0.05, so the homogeneity of variances assumption is met

# runs vs fitted values plot

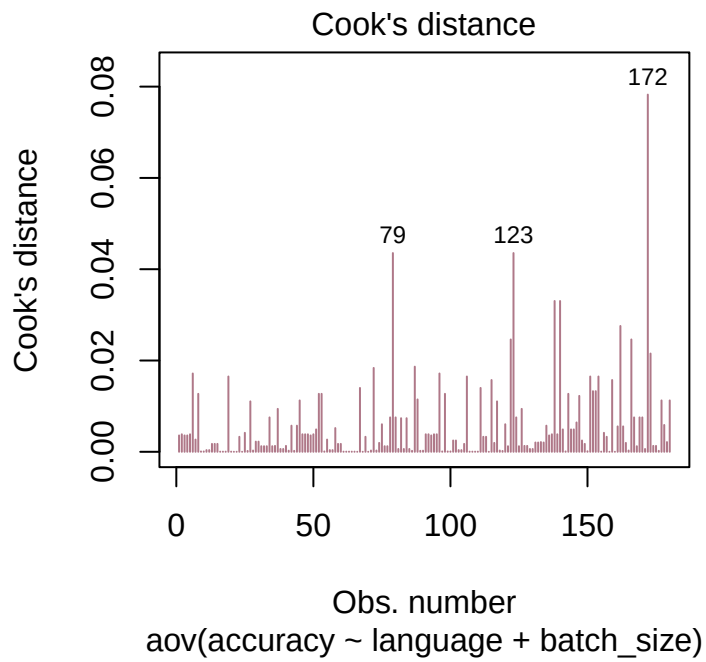
plot(fit5, 1,
     col = adjustcolor("#7A1E3A", 0.6),
     pch = 16)
plot(fit5, 2, col = adjustcolor("#7A1E3A", 0.6), pch = 16) # normal Q-Q plot
```



```
plot(fit5, 3, col = adjustcolor("#7A1E3A", 0.6), pch = 16 ) # scale-location plot
```

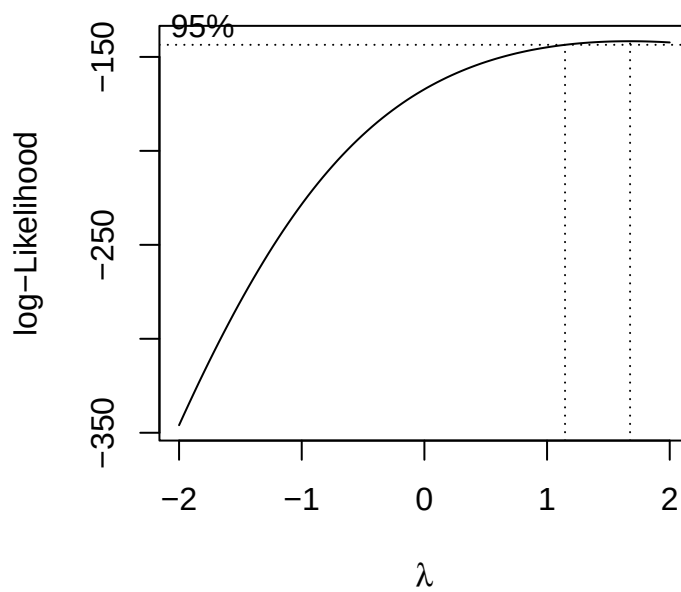


```
plot(fit5, 4, col = adjustcolor("#7A1E3A", 0.6), pch = 16 ) # residuals vs leverage plot
```



```
# using boxcox transformation to meet the normality assumption
```

```
boxcox(fit5, lambda = seq(-2, 2, 0.1), main = "Box-Cox Transformation for accuracy")
```



```
# lambda = approx. 1.6 is the best transformation
```

```
# with the boxcox transformation, we can transform the response variable and fit the model again
data$accuracy_transformed <- data$accuracy^1.6
```

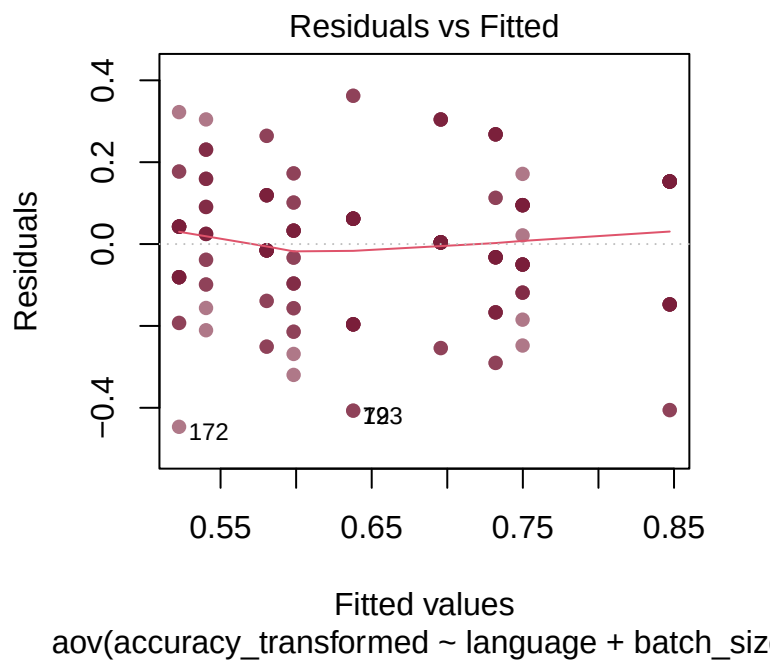
```
fit6 <- aov(accuracy_transformed ~ language + batch_size, data = data)
summary(fit6)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
language	2	1.403	0.7015	25.558	1.83e-10 ***
batch_size	2	0.461	0.2307	8.403	0.000328 ***
Residuals	175	4.803	0.0274		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Assumptions ~ Normality, Homogeneity of variances, Independence of residuals, Randomness of residuals, No outliers or influential
```

```
plot(fit6, 1,
     col = adjustcolor("#7A1E3A", 0.6),
     pch = 16) # residuals vs fitted values
```



```
leveneTest(accuracy_transformed ~ language, data = data)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  0.4648  0.629
177
```

```
leveneTest(accuracy_transformed ~ batch_size, data = data)
```

```

Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  0.8911 0.4121
      177

```

```

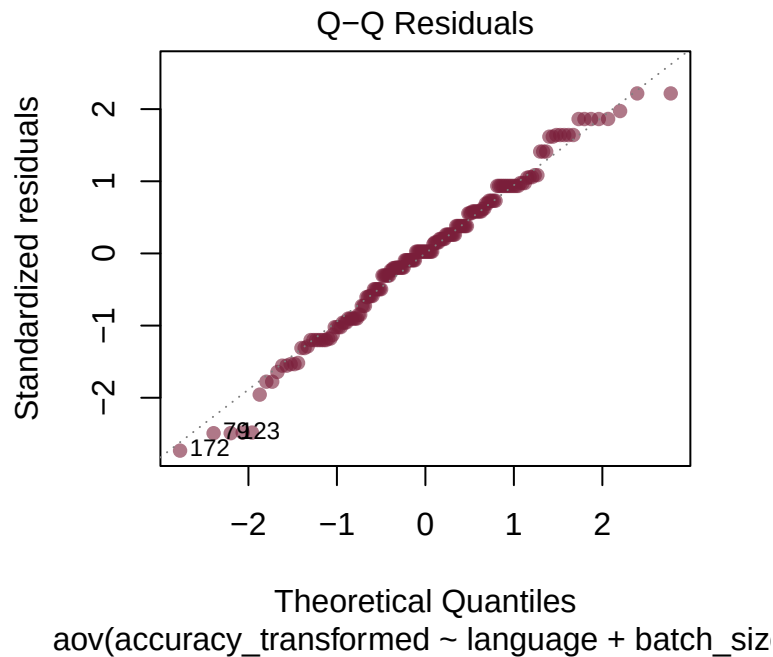
# p-value ~ 0.629 and 0.4121 resp., so the homogeneity of variances assumption is met

```

```

plot(fit6, 2, col = adjustcolor("#7A1E3A", 0.6), pch = 16 ) # normal Q-Q plot

```



```

shapiro.test(residuals(fit6))

```

Shapiro-Wilk normality test

```

data: residuals(fit6)
W = 0.9873, p-value = 0.1049

```

```

# p-value ~ 0.1049, so the normality assumption is met

```

```

# runs test

```

```

runs.test(residuals(fit6))

```

Runs Test

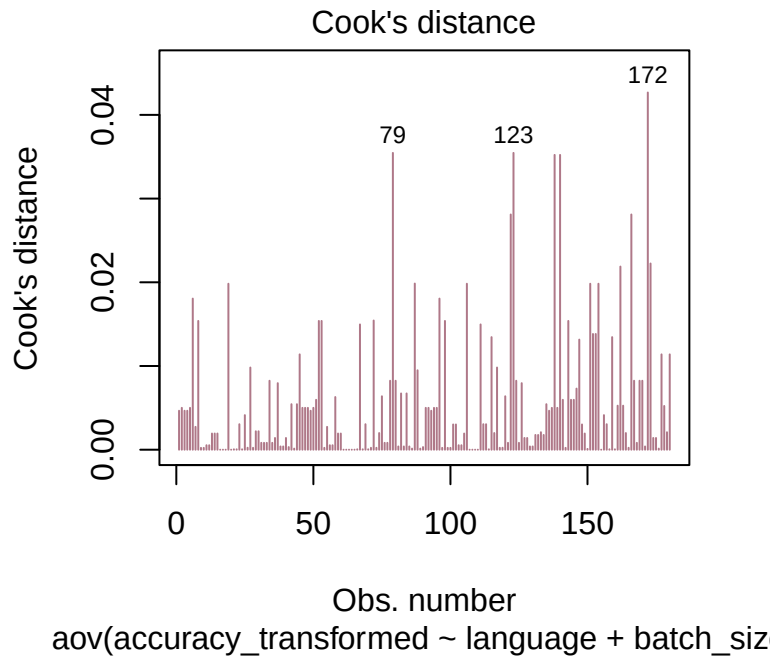
```

data: residuals(fit6)
statistic = -0.30957, runs = 82, n1 = 84, n2 = 82, n = 166, p-value =
0.7569
alternative hypothesis: nonrandomness

```

```
# p-value is 0.7569, so the residuals are random

# outliers and influential points
plot(fit6, 4, col = adjustcolor("#7A1E3A", 0.6), pch = 16) # residuals vs leverage plot
```



```
# there are outliers and influential points flagged
# but these are valid data points
# so we will keep them in the analysis

# post-hoc test for language
TukeyHSD(fit6, "language")
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = accuracy_transformed ~ language + batch_size, data = data)
```

```
$language
      diff      lwr      upr    p adj
cn-ar 0.05794729 -0.01355371 0.1294483 0.1372558
en-ar 0.20941050  0.13790950 0.2809115 0.0000000
en-cn 0.15146322  0.07996222 0.2229642 0.0000040
```

```
TukeyHSD(fit6, "batch_size")
```

Tukey multiple comparisons of means
95% family-wise confidence level

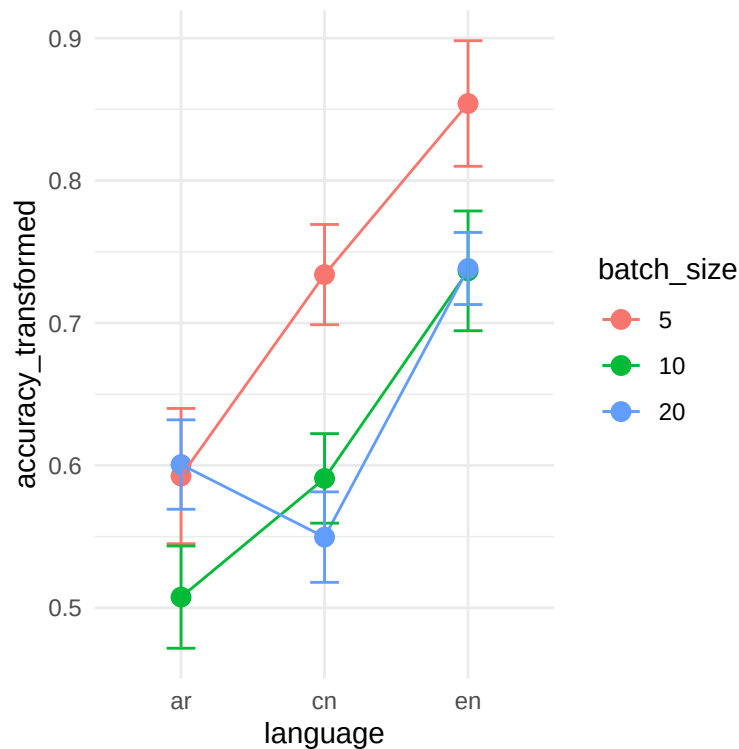
```
Fit: aov(formula = accuracy_transformed ~ language + batch_size, data = data)
```

```
$batch_size
      diff      lwr      upr    p adj
10-5  -0.11519341 -0.1866944 -0.04369241 0.0005647
```

```
20-5 -0.09735451 -0.1688555 -0.02585351 0.0043601
20-10 0.01783890 -0.0536621 0.08933990 0.8257963
```

```
# plot the interaction between language and batch size
```

```
ggplot(data, aes(x = language, y = accuracy_transformed, color = batch_size, group = batch_size)) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun = mean, geom = "line") +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2) +
  theme_minimal()
```



```
# Now language and model
fit7 <- aov(accuracy ~ language + model, data = data)
summary(fit7)
```

```
          Df Sum Sq Mean Sq F value    Pr(>F)
language    2  0.7676   0.3838   23.131 1.24e-09 ***
model        3  0.1174   0.0391    2.359  0.0732 .
Residuals  174  2.8870   0.0166
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# model is not significant at 0.05, but language is still significant
```

```
# Model and batch size
fit8 <- aov(accuracy ~ model + batch_size, data = data)
summary(fit8)
```

```
          Df Sum Sq Mean Sq F value    Pr(>F)
model        3  0.117 0.03915    1.991 0.11710
batch_size    2  0.233 0.11667    5.934 0.00321 **
Residuals  174  3.421 0.01966
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# batch size is significant, but model is not significant at 0.05 level
```

```
# three-way ANOVA for language, model and batch size
fit9 <- aov(accuracy ~ language * model * batch_size, data = data)
summary(fit9)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
language	2	0.7676	0.3838	23.914	1.08e-09 ***
model	3	0.1174	0.0391	2.439	0.066887 .
batch_size	2	0.2333	0.1167	7.270	0.000982 ***
language:model	6	0.0513	0.0086	0.533	0.782641
language:batch_size	4	0.1228	0.0307	1.913	0.111325
model:batch_size	6	0.0789	0.0131	0.819	0.556673
language:model:batch_size	12	0.0896	0.0075	0.465	0.931960
Residuals	144	2.3110	0.0160		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# further confirms only language and batch size are significant, but not model or interactions
```

```
# fit6 is the best model, so we will use it will be the final fit
```